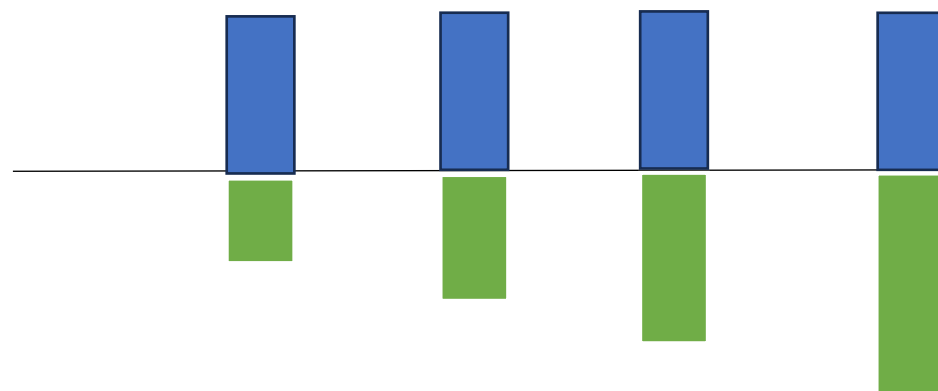


RFR Regime (3. SOFR Swaps)

Fixed-Floating Swap Basics

- ▶ One party pays **fixed coupons** whereas another party pays **floating coupons** tied to a benchmark
- ▶ The floating payment may or may not include an additional spread
- ▶ The fixed rate that makes the value of the swap = 0 on any day is the **swap rate** or **par rate**.
Par rates can be considered as average of the floating rates over the life of the instrument.
- ▶ Long on swap = long on floating leg = fixed payer = payer swap
- ▶ Short on swap = short on floating leg = fixed receiver = receiver swap



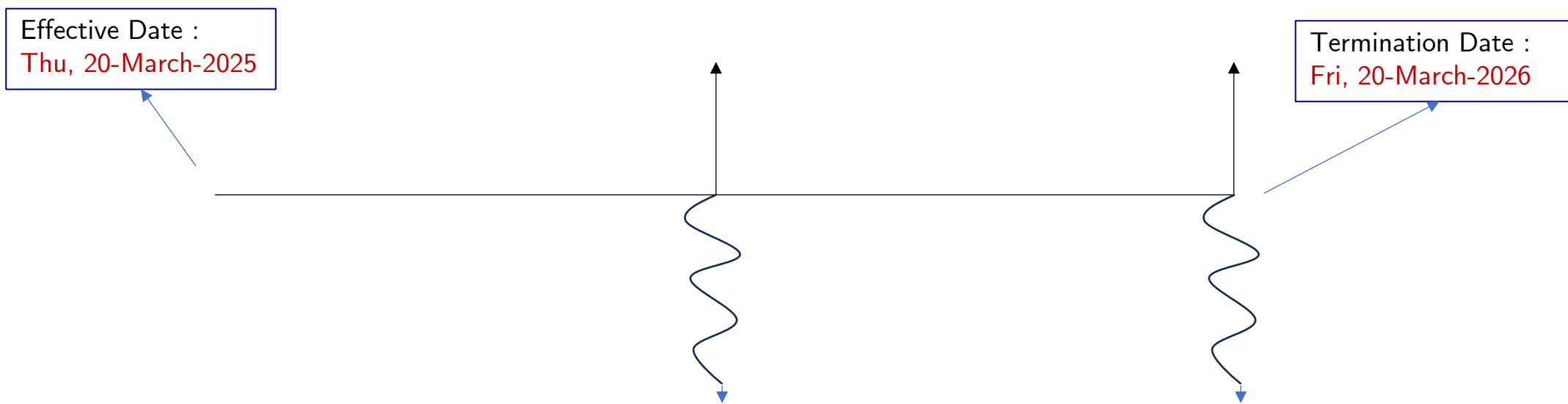
(typical cashflow profile for a receiver swap under an upward sloping yield curve)

USD SOFR Swap Template

Swap	Leg1 : FIXED	Leg2 : FLOAT
Pay/Receive	RECEIVE	PAY
Notional	100,000,000	100,000,000
Currency	USD	USD
Effective Date	0D Thu, 20-March-2025	0D Thu, 20-March-2025
Termination Date	1Y Fri, 20-March-2026	1Y Fri, 20-March-2026
Fixed Rate (%)	3.4%	
Float Index		1D USD SOFR
Float Spread (bps)		0.00
Reset Frequency		DAILY
Payment Frequency	SEMI-ANNUALLY	QUARTERLY
Day Count Convention	30/360	ACT/360

Market	Leg1 : FIXED	Leg2 : FLOAT
Curve Date	0D Thu, 20-March-2025	0D Thu, 20-March-2025
Forecast Curve		USD SOFR
Discount Curve	USD SOFR	USD SOFR

Effective Date & Termination Date



- ▶ **Effective Date** : Start date of the swap
- ▶ **Termination Date** : End date (Maturity of the swap)

Calendar



- ▶ A calendar is a set $\mathcal{C}$ of **non-business** days or holidays . Examples of calendars are TARGET(Trans-European Automated Real-time Gross settlement Express Transfer), LONDON, NYC etc.
- ▶ A day d is non-business day if $d \in \mathcal{C}$
- ▶ Calendars need to be specified in the context of a financial product



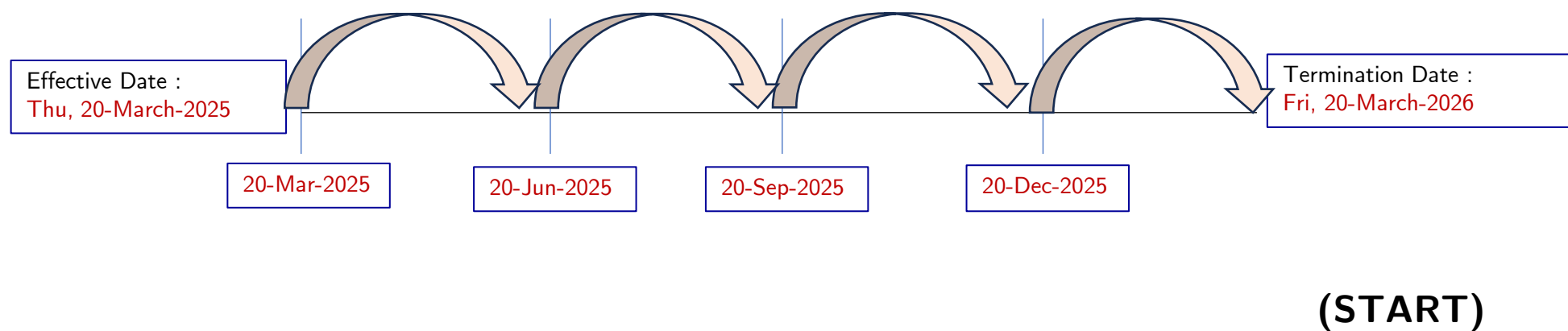
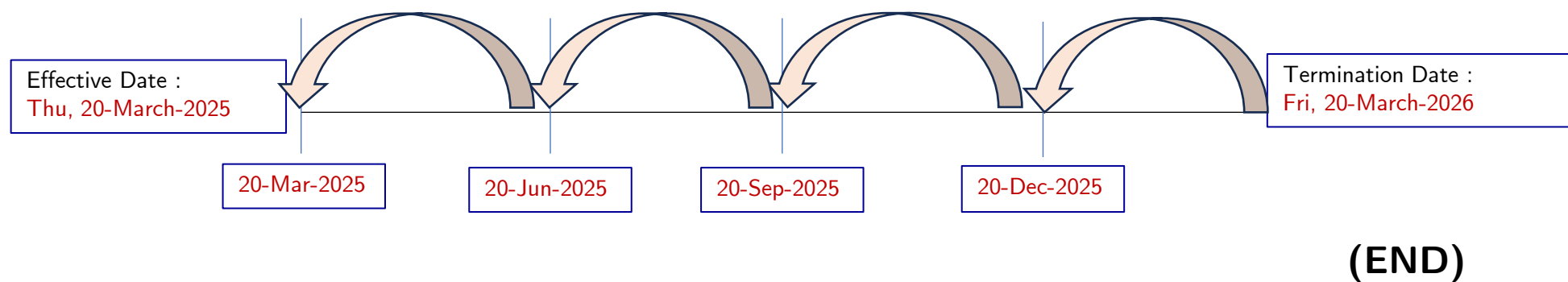
Roll Day



- **Roll day** is used to determine the unadjusted coupon accrual end dates. It can be specified either by a number between 1-31 or one of the values in the list { IMM, EOM, START, END }

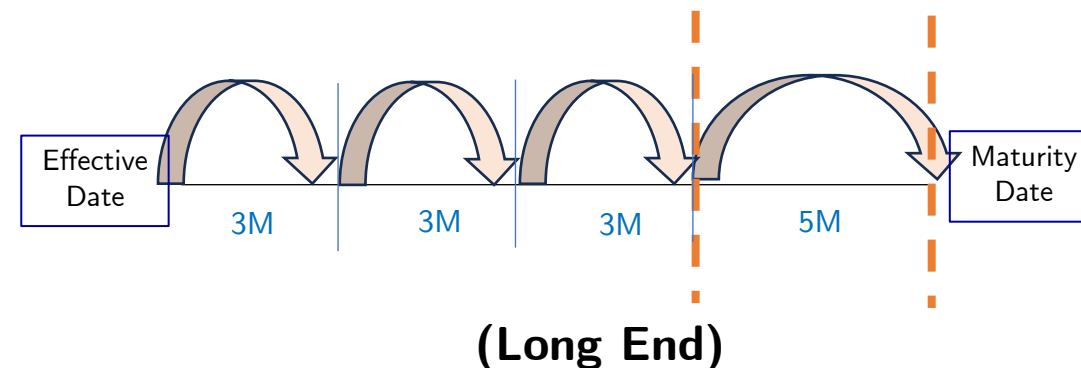
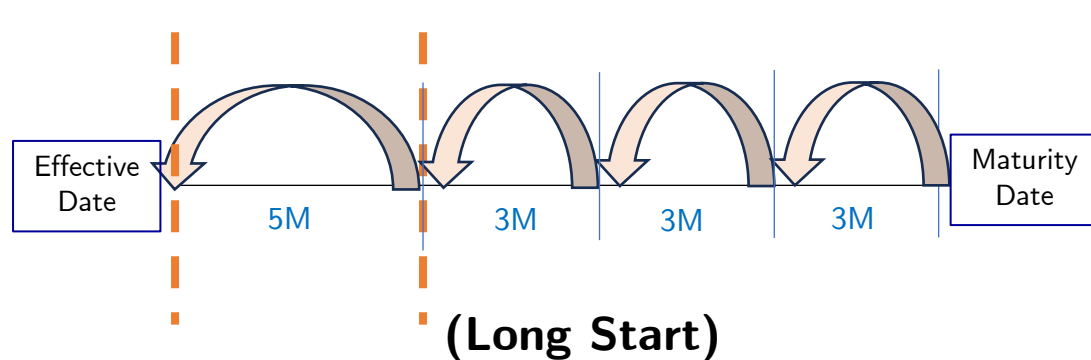
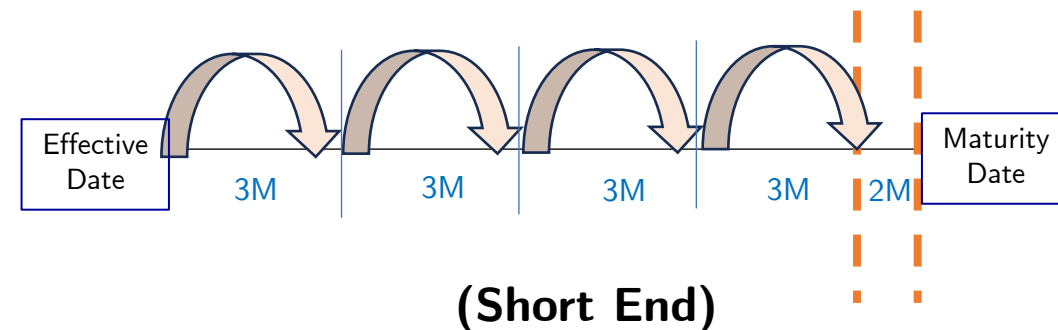
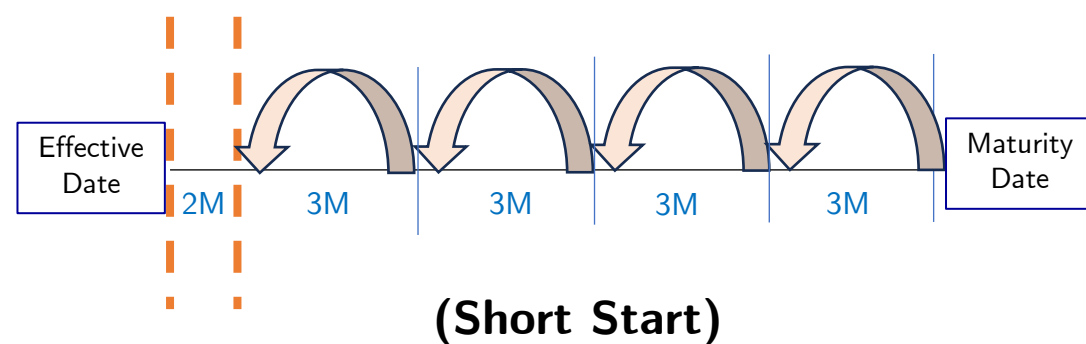
Roll Day	Method
1-31	The coupon accrual end date falls on this day of the month
IMM	Accrual end date is an IMM date i.e., 3 rd Wednesday of March, June, September or December
EOM	Coupons accrue to the end of the month
START	Coupons accrue to the same day as the unadjusted effective date
END (default)	Coupons accrue to the same day as the unadjusted termination date

Roll Day - Examples



Stub type

- ▶ For swaps of irregular length, typically there will be one coupon period which will be shorter or longer than others which are referred to as stubs. For example, consider a swap length of 1Y 2months with quarterly payment frequency. We have the following variants



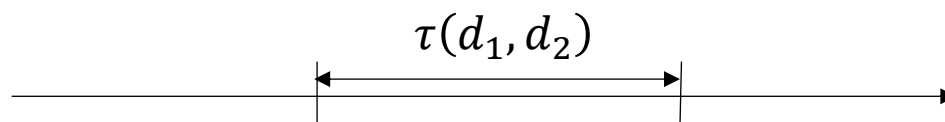
Day count convention

- ▶ Day count convention is used to calculate the year fraction between 2 dates which is used to calculate accrued coupon amount. It's a function $\tau : \mathcal{D} \times \mathcal{D} \rightarrow \mathbb{R}$

- ▶ **Act / 365** : $\tau(d_1, d_2) = (d_2 - d_1)/365$

- ▶ **Act / 360** : $\tau(d_1, d_2) = (d_2 - d_1)/360$

- ▶ **30 / 360** :
$$\tau(d_1, d_2) = \frac{(\max(30 - d_1, 0) + \min(d_2, 30) + 360 \times (y_2 - y_1) + 30 \times (m_2 - m_1))}{360}$$



Business Day Convention

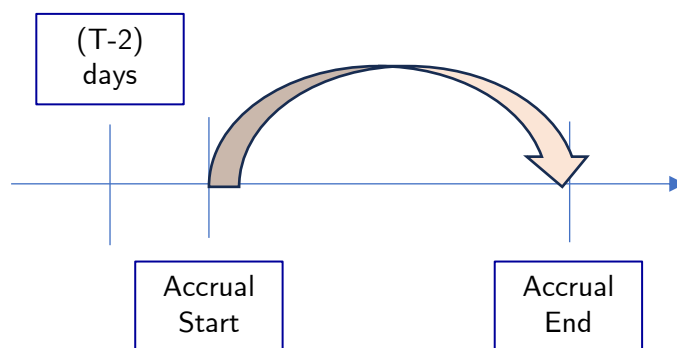


- ▶ A business day convention is a function $\omega_{\mathcal{C}} : \mathcal{D} \rightarrow \mathcal{D}$ that maps a date to another date given a calendar $\mathcal{C}$. Business days remain unchanged i.e., $\omega_{\mathcal{C}}(d) = d$ if $d \notin \mathcal{C}$
- ▶ It's a means to adjust the accrual/payment days to business days if they fall on a holiday. We have the following conventions

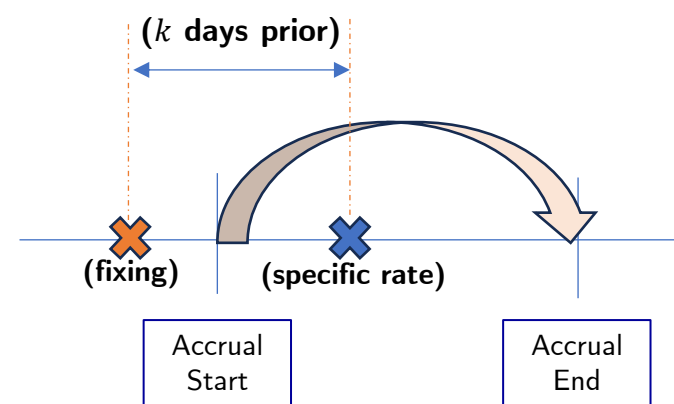
Business Day Convention	Method
FOLLOWING	Adjust to the next business day $\omega_{\mathcal{C}}(d) = \min \{ \tilde{d} \mid \tilde{d} \notin \mathcal{C}, \tilde{d} \geq d \}$
PRECEDING	Adjust to the previous business day $\omega_{\mathcal{C}}(d) = \max \{ \tilde{d} \mid \tilde{d} \notin \mathcal{C}, \tilde{d} \leq d \}$
MODIFIED FOLLOWING	Adjust to next business day, unless it falls on the next month in which case adjust to the previous business day
MODIFIED PRECEDING	Adjust to previous business day, unless it falls on the previous month in which case adjust to the next business day
UNADJUSTED	No Change

Fixing Date

- ▶ Floating Coupon payment over an accrual period is calculated by either averaging or compounding rates over that period. But it's not necessary to use the rates observed exactly during this period. Typically, rates used in the calculation are borrowed from a few days prior to for operational convenience.
- ▶ In LIBOR days the floating rate was fixed 2 days prior to the start of the period. In RFR regime there are multiple conventions available. (More on later slides)



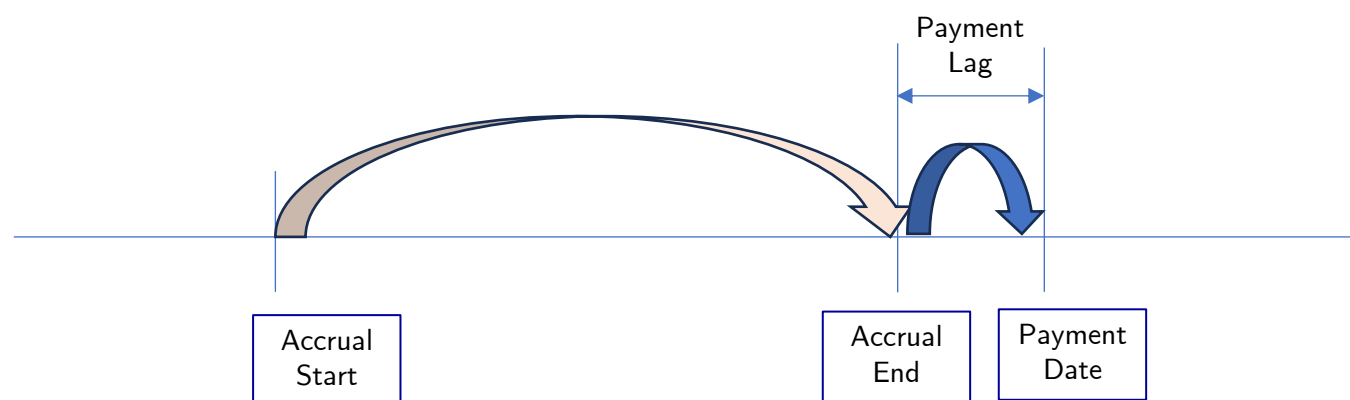
LIBOR



RFR

Payment Lag

- ▶ A **payment lag** simply postpones the payment after the accrual end. This is again to improve operational efficiency. But more importantly, the discount factor for the present value calculation need to consider the payment lag days. Typically, it's 2 business days post holiday-adjusted accrual end date.



ISDA – RFR based Floating rate standard



- ▶ Given the backward-looking nature of RFR, the simple settlement in arrears standard is not practical as the full rate becomes known only hours before the payment. To overcome this practical issue, ISDA proposed a modular approach using overnight RFR rates for compounding/averaging

- ☐ Observation shift
- ☐ Lookback
- ☐ Lockout

ISDA – RFR based Floating rate standard



(Accrual Period 1)

(Accrual Period 2)

	Day 1	Day 2	...	Day (T-2)	Day (T-1)	Day T	Day (T+1)	Day (T+2)
Publication		SOFR for Day 1	...	SOFR for Day (T-3)	SOFR for Day (T-2)	SOFR for Day (T-1)	SOFR for Day (T)	
Plain Arrears	Use SOFR for Day 1	Use SOFR for Day 2	...	Use SOFR for Day (T-2)	Use SOFR for Day (T-1)	Use SOFR for Day (T)	Payment Due	
Arrears with Payment Delay	Use SOFR for Day 1	Use SOFR for Day 2	...	Use SOFR for Day (T-2)	Use SOFR for Day (T-1)	Use SOFR for Day (T)		Payment Due
Arrears with 1-day Lockout	Use SOFR for Day 1	Use SOFR for Day 2	...	Use SOFR for Day (T-2)	Use SOFR for Day (T-1)	Use SOFR for Day (T-1)	Payment Due	
Arrears with 1-day Lookback	Use SOFR for Day 0	Use SOFR for Day 1	...	Use SOFR for Day (T-3)	Use SOFR for Day (T-2)	Use SOFR for Day (T-1)	Payment Due	

Lookback variants

- ▶ Let's say there is a rate r_i in the accrual period which needs to be included in the compounding and the applicable # of calendar days for r_i is d_i . Under Lookback, we use a rate observed k -days prior i.e., r_{i-k} . Lookback period is typically between 2-5 days. But there are couple of important ways in which lookback can be enforced.
- ▶ **Lookback without observation shift** – We borrow only the past rate r_{i-k} but NOT the day count from the past. The applicable days for the rate to be used in compounding would be still d_i
- ▶ **Lookback with observation shift** – We borrow both the past rate, and it's associated calendar days to use in compounding.

Lookback variants

(Example : 5-days lookback)

Date	d_i	SOFR
Mon, Jun 24, 2019	1	2.39
Tue, Jun 25, 2019	1	2.41
Wed, Jun 26, 2019	1	2.43
Thu, Jun 27, 2019	1	2.42
Fri, Jun 28, 2019	3	2.5
Mon, Jul 1, 2019	1	2.42
Tue, Jul 2, 2019	1	2.51
Wed, Jul 3, 2019	2	2.56
Fri, Jul 5, 2019	3	2.59

(without observation shift)

Date	d_i	SOFR
Mon, Jun 24, 2019	1	2.39
Tue, Jun 25, 2019	1	2.41
Wed, Jun 26, 2019	1	2.43
Thu, Jun 27, 2019	1	2.42
Fri, Jun 28, 2019	3	2.5
Mon, Jul 1, 2019	1	2.42
Tue, Jul 2, 2019	1	2.51
Wed, Jul 3, 2019	2	2.56
Fri, Jul 5, 2019	3	2.59

(with observation shift)

Swap Schedule generation

